MIL-C- 13613 (Ord)
3 September 1954
SUPERSEDING
PA-PD-185 (Rev 1)
27 April 1954

MILITARY SPECIFICATION

CARBAZOLE

1. SCOPE

1.1 This specification covers carbazole for use in the manufacture of tetranitrocarbazole (TNC).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, and all specifications, and drawings referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 -General Specification For Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-129 -Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Purity.-The purity shall be 97.0 percent, minimum, when determined as specified in 4.4.

3.2 Freezing point.-The freezing point shall be 244.0°C, minimum, when determined as specified in 4.5.

3.3 Ash content.-The ash content shall be 0.10 percent, maximum, when determined as specified in 4.6.

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3.4 Moisture content.-The moisture content shall be 0.10 percent, maximum, when determined as specified in 4.7.

3.5 Color.-The color shall be white to grayish-white, when determined as specified in 4.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Lot.-A lot shall consist of material from batches which have been manufactured by the same process by one manufacturer.

A batch shall be defined as that quantity of material which has been subjected to some unit chemical or physical mixing operation which serves to make the product substantially uniform.

4.2 Sampling.-A sample of approximately 1 pound shall be taken from one of the containers in the lot. This sample shall be placed in a rubber stoppered bottle labeled to show the name of the material, manufacturer, plant, contract or order number, lot number, and lot size. All acceptance tests shall be made on this sample.

4.3 Inspection.-Inspection shall be made in accordance with Specification MIL-G-2550.

4.4 Determination of purity.-An accurately weighed portion of 2.0 ± 0.5 gm of the sample shall be transferred to an 800-ml Kjeladhl digestion flask. Ten gm of powdered potassium sulfate, 0.3 gm of crystallized copper sulfate, and 50 ml of concentrated sulfuric acid shall be added. Several glass beads shall be added to the flask to avoid bumping. The flask shall be placed in an inclined position and heated below the boiling point of the acid until frothing ceases. The heat shall be increased until the acid boils briskly, and digested for 4 hours. The contents of the flask shall be cooled to room temperature and diluted with 200 ml of water. A 100-ml portion of a 50-percent sodium hydroxide solution (equal weights of sodium hydroxide and water) shall be added slowly. This sodium hydroxide solution shall be poured down the side of the flask so that it does not mix with the acid solution. The Kjeldahl flask shall be connected to a condenser by means of a Kjeldahl connecting bulb and the tip of the condenser shall be extended below the surface of a 100-ml solution of 4-percent boric acid contained in a 500-ml Erlenmeyer flask. The contents of the Kjeldahl flask shall be mixed by shaking. Approximately 150 ml of the liquid shall be distilled into the boric acid solution. This new solution shall be titrated with 0.3N hydrochloric acid, using methyl red indicator. Correction for a blank determination shall be made, and the purity of the carbazole calculated as follows:

$$\text{Purity, percent} = \frac{16.72\ AN}{W}$$

where:

A = ml of hydrochloric acid used in sample titration minus ml of hydrochloric acid used in blank titration.
N = Normality of hydrochloric acid.
W = Weight of the sample on a moisture-free basis.

4.5 Determination of freezing point.-The equipment (see figure 1) shall be assembled with the inner test tube, thermometers, and stirrer

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withdrawn to allow the carbazole to be placed in the test tube and melted with the aid of a Bunsen burner. The bulb of the correction thermometer (with 1°C subdivisions) shall be about 1 cm from and adjacent to the middle of the exposed mercury column of the thermometer immersed in the sample. Sufficient carbazole shall be placed in the test tube to cover the standardized thermometer (calibrated for total immersion and having 0.2°C sub-divisions) to at least 3 inches above the bulb. Shortly after the thermometer is adjusted, the carbazole shall be completely melted again, care being taken not to heat the carbazole over 260°C, and the smaller tube shall be inserted into the larger one. The carbazole shall be agitated vigorously with the metal stirrer as it cools. As the temperature lowers, the thermometer shall be read every 1/2 minute until the temperature remains constant, or practically so, and then begins to drop again. Occasionally supercooling may occur which will cause the temperature to drop below the freezing point, and then rise to the freezing point. Should supercooling occur, the highest temperature obtained after the temperature starts to rise shall be taken as the uncorrected freezing point. Should no supercooling occur, the temperature shall be recorded as the uncorrected freezing point when it becomes constant, or nearly so, during the freezing of the material. The reading of the correction thermometer shall be recorded, and considered the average temperature of the exposed mercury column. The emerged stem correction shall be calculated as follows:

$$\text{Correction (degrees C)} = N(T - t)\ (0.00016)$$

where:

N = Number of degrees in the exposed mercury column.
T = Uncorrected freezing point.
t = Average temperature of the exposed mercury column.

The corrected freezing point shall be calculated by adding this correction to the uncorrected freezing point.

4.6 Determination of ash content.-An accurately weighed portion of approximately 10 gm of the sample shall be transferred to a tared porcelain or platinum crucible. The crucible and contents shall be ignited by means of a gas flame or ruffle furnace until all combustible matter has been removed, cooled in a desiccator, and weighed. The weight of the residue shall be calculated as percentage of ash.

4.7 Determination of moisture content.-An accurately weighed portion of approximately 5 gm of the sample shall be transferred to a tared moisture dish. The dish and contents shall be placed in an oven maintained at 100° to 105°C for 2 hours, cooled in a desiccator, and weighed. The loss in weight of the dish and contents shall be calculated to percent moisture in the sample as follows:

$$\text{Percent moisture} = \frac{100A}{W}$$

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where:

A = Loss in weight of moisture dish and contents.
W = Weight of the sample.

4.8 Color.-The color shall be determined by visual inspection.

4.9 Retest.-Should a lot fail to pass the specified tests, it shall be rejected, but it may, upon application from the manufacturer and approval of the contracting officer, be retested once.

5. PREPARATION FOR DELIVERY

5.1 Packing.-Unless otherwise specified, carbazole shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

5.2 Marking.-In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

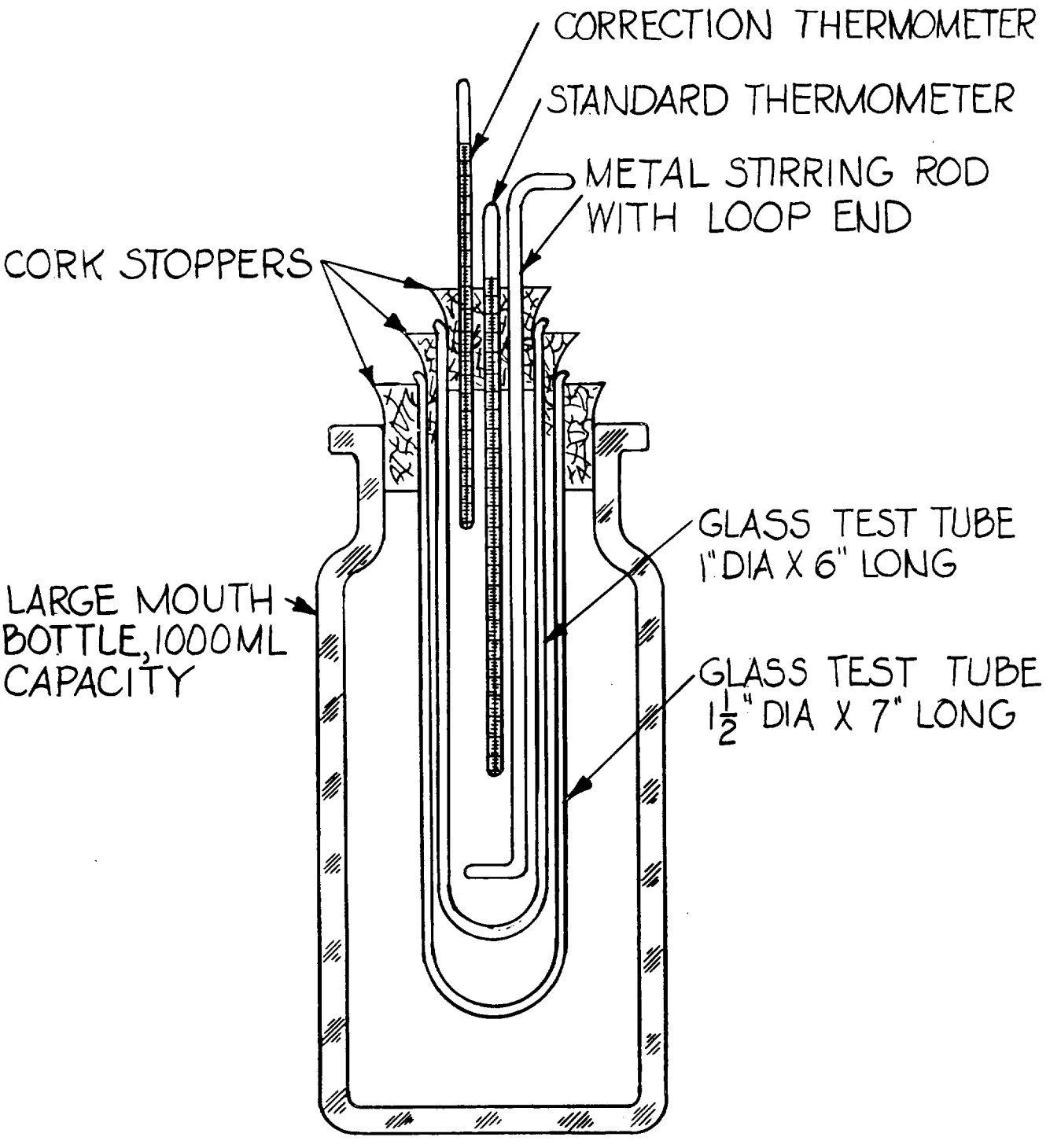
6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

Notice.-This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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SOLIDIFICATION POINT APPARATUS

FIGURE 1

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